



# Transit-Oriented Development-Based Strategies for Enhancing Commercial Spatial Vitality in Changchun's North Ring Road Station Area

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**Abstract:** Transit-oriented development (TOD) offers a planning basis for compact, mixed-use, and vibrant station areas. However, in many rapidly developing Chinese cities, metro-network expansion has not been matched by coordinated station-area commercial renewal, resulting in spatially uneven street-level activity despite improved transit accessibility. Taking the North Ring Road Station area in Changchun, China, as a case, this study examines the associations between TOD-related spatial conditions and observed human activity intensity as a proxy for commercial spatial vitality, and how these spatial associations can inform targeted regeneration strategies. Within an 800 m TOD analytical boundary, commercial point of interest (POI) data, Baidu Heatmap-derived human activity data, building footprints, road-network data, land-use information, and field observations were integrated. Average nearest-neighbor (ANN) analysis, kernel density estimation (KDE), standard deviational ellipse (SDE) analysis, ordinary least squares (OLS) regression, and geographically weighted regression (GWR) were applied within a 5D TOD framework. The results reveal a dual-core, corridor-oriented, and spatially uneven commercial structure. Commercial facility agglomeration and functional diversity are positively associated with observed activity intensity, whereas transit distance, road-network configuration, building morphology, and proximity to commercial anchors exhibit spatially heterogeneous associations. These results suggest that transit proximity and commercial concentration alone do not necessarily correspond to spatially continuous activity when pedestrian permeability, interface openness, and functional mixing are limited. The study translates these associations into evidence-informed transport-development responses that coordinate metro-entrance access, pedestrian transfer and first/last-mile connections, commercial functions, and surrounding land uses, thereby providing a micro-scale diagnostic approach to station–street–commercial integration.

**Keywords:** Transit-oriented development; Commercial spatial vitality; Metro station; Point of interest; Station area; Spatial heterogeneity; Geographically weighted regression

## 1 Introduction

Transit-oriented development (TOD) is used to guide compact, mixed-use, and pedestrian-oriented station area development. By clustering people, services, and activities around transit nodes, TOD is expected to enhance accessibility and urban vitality, as discussed by Calthorpe [1] and Cervero and Kockelman [2]. Metro station areas are therefore key sites for commercial activity, public life, and TOD-based regeneration.

However, rail investment does not necessarily generate balanced commercial vitality. In many rapidly developing Chinese cities, metro construction has outpaced the coordinated development of station area land use, public space, pedestrian environments, and commercial functions. Many station areas remain spatially uneven, with strong cores, corridor dependence, and limited spillover into adjacent areas [3]. For TOD practice, this raises the question of how transit accessibility is associated with commercial vitality under varying spatial configurations and local development conditions.

Although the TOD 5D framework has advanced station-area research through density, diversity, design, destination accessibility, and distance to transit, as discussed by Ewing and Cervero [4] and Knowles et al. [5], existing studies often emphasize broader relationships between built-environment conditions and urban vitality. As discussed by Chen et al. [6] and Jin et al. [7], multisource data have improved the measurement of urban activity, while point of interest (POI) and heatmap data allow activity patterns to be examined at finer spatial scales. Recent TOD–vibrancy studies have demonstrated the value of multisource urban data, TOD-related built-environment indicators, and nonlinear or spatially differentiated explanatory models at city, district, and multi-station scales. Li et al. [8] integrated multiple built-environment dimensions with multisource vitality data, Yu et al. [9] examined the spatiotemporal relationships between TOD and vibrancy, and Xiao et al. [10] used interpretable machine learning to identify nonlinear and synergistic TOD–vibrancy relationships. These studies indicate that multisource data integration and the application of local or nonlinear models are not, by themselves, methodological novelties. A remaining gap concerns the fine-scale diagnosis of heterogeneity within a single station area and, in particular, how local statistical associations can be translated into place-specific spatial diagnosis and regeneration responses. Accordingly, the methodological advance of this study lies in linking 30 m grid-based activity measurement, TOD 5D indicators, global–local association analysis, and field-observed spatial conditions within a common intra-station diagnosis-to-strategy framework.

To address this gap, this study examines the North Ring Road Station area in Changchun, an interchange station serving Rail Transit Lines 1 and 8. It combines POI data, Baidu Heatmap data, spatial analysis, and the TOD 5D framework to examine commercial structure and observed human activity intensity at the micro scale. Average nearest-neighbor (ANN), kernel density estimation (KDE), and standard deviational ellipse (SDE) analyses are first used to describe the spatial organization of commercial facilities, followed by ordinary least squares (OLS) and geographically weighted regression (GWR) to identify global and spatially varying associations between TOD-related conditions and observed activity intensity, following Clark and Evans [11] and Fotheringham [12]. These statistical patterns are subsequently interpreted together with field observations to support spatial diagnosis and the translation of findings into planning responses.

This study contributes to TOD-based station-area research in three respects. First, it shifts the analytical focus from inter-station or aggregate station-level comparison to fine-scale intra-station heterogeneity using a common 30 m analytical grid. Second, rather than treating GWR itself as a methodological novelty, the study uses spatially varying coefficients as diagnostic evidence and interprets them together with field-observed pedestrian, interface, and land-use conditions. Third, it connects fine-scale spatial diagnosis with transport-development integration by examining how metro access, pedestrian movement structure, commercial destinations, and surrounding land uses jointly correspond to observed activity patterns, and by translating these relationships into differentiated station–street–commercial regeneration responses.

This paper is organized as follows. Section 2 reviews the relevant literature on TOD, commercial spatial vitality, built-environment heterogeneity, and justice in metro and TOD development. Section 3 introduces the study area, data sources, and analytical methods. Sections 4 and 5 examine the commercial structure, observed activity patterns, and TOD-related explanatory variables. Section 6 presents the global and spatially varying associations identified by the OLS and GWR analyses. Section 7 discusses the main findings, justice and stakeholder implications, evidence-informed spatial diagnosis, and TOD-based planning strategies. Section 8 concludes the paper by summarizing the main findings, limitations, and broader applicability of the proposed benchmarking framework.

## 2 Literature Review

### 2.1 Transit-Oriented Development and Commercial Spatial Vitality in Station Areas

TOD integrates transit investment with surrounding land-use development by promoting compact form, mixed functions, and walkable station areas. Early TOD studies focused on density, diversity, and design. Later studies added destination accessibility and distance to transit. These five dimensions formed the 5D framework and further connected TOD research with node-place analysis, as discussed by Knowles et al. [5], Bertolini [13], and Papa and Bertolini [14]. Observed human activity represents an observable dimension of commercial spatial vitality that allows TOD-related spatial conditions to be examined at the station-area level. From a TOD perspective, rail transit provides accessibility and potential movement opportunities around stations; however, the extent to which these are translated into street-level human activity may vary with multiple interacting spatial and functional conditions, including pedestrian connectivity, commercial functions, public space, and street interfaces, as discussed by Yu and Lu [15]. The surrounding environment of subway stations, such as street continuity, functional mix, and the openness of commercial interfaces, influences whether people stay, move through the area, or disperse after exiting the station.

Commercial buildings also influence commercial vitality within station areas. Their spatial organization and degree of openness affect whether pedestrian flows can extend from commercial buildings into surrounding streets. Enclosed commercial complexes provide a typical example. They can attract large flows, but much of the activity remains inside the buildings. As a result, nearby streets, small street-level businesses, and public spaces may receive limited spillover from metro accessibility and commercial agglomeration, as discussed by Dovey and Pafka [16].

More broadly, railway infrastructure is closely associated with regional economic and social development [17]. In the present station-area context, this broader relationship highlights the need to consider transport infrastructure together with the surrounding built environment and the connections between commercial buildings, streets, and public spaces.

## **2.2 Measurement of Commercial Spatial Vitality and Observed Human Activity**

The measurement of commercial vitality in station areas should begin with spatial vitality, focusing on the aggregation, dwelling, and diffusion characteristics of crowd activity in the surrounding space. Simultaneously, commercial vitality also depends on factors such as commercial supply, functional mixing, street continuity, spatial accessibility, and the quality of public spaces. Jacobs [18], Montgomery [19], and Gehl [20] identified mixed-use development, continuous street life, and high-quality public spaces as important foundations for maintaining urban vitality. Currently, digital trace data provide new approaches for measuring urban activity. As discussed by Jin et al. [7], Kim [21], Sulis et al. [22], and Tu et al. [23], data such as mobile phone signaling and heat maps can reflect crowd aggregation and activity distribution with high spatiotemporal resolution. In this study, commercial spatial vitality refers to the capacity of station-area commercial spaces and surrounding environments to attract and sustain human activity. Baidu Heatmap-derived activity intensity is used as a spatial proxy for its observable activity dimension, while commercial POIs, functional diversity, built-environment indicators, and field observations provide contextual information for interpreting the observed spatial patterns. The heatmap data reflect relative human presence and do not directly measure commercial transactions or consumer behavior.

## **2.3 Built-Environment Factors and Spatial Heterogeneity**

Built-environment conditions are closely associated with commercial spatial vitality in station areas. Commercial facility distribution, building morphology, road-network structure, functional mixing, and locational accessibility provide different spatial conditions for pedestrian activity and the use of commercial space. Their relationships with observed activity may also vary according to local connectivity, interface conditions, and functional composition, as discussed by Gao et al. [24].

The associations between built-environment conditions and commercial spatial vitality may vary across space. As discussed by Brunsdon et al. [25] and Wang et al. [26], GWR can reveal local spatial relationships that are not fully captured by global models. Existing research largely focuses on activity hotspots or broader station-level relationships, while fine-scale spatial differences in how TOD-related conditions correspond to activity within a single station area remain less examined. Following Li et al. [8], this study combines Baidu Heatmap data, the TOD 5D framework, and OLS/GWR to examine global and spatially varying associations between observed activity intensity and TOD-related spatial conditions. Field observations are subsequently used to provide contextual interpretations of these statistical relationships. The estimated relationships are interpreted as spatial associations rather than causal effects.

## **2.4 Justice in Metro and Transit-Oriented Development**

Beyond accessibility and urban vitality, metro and TOD development also raises questions of spatial and process justice. Spatial justice concerns how the benefits and constraints associated with transport accessibility and the surrounding urban environment are distributed among different users and places. Tzanni et al. [27] showed that deficiencies in pedestrian infrastructure and longer walking distances can discourage metro use, while socioeconomic conditions may influence people's capacity to benefit from transport-related improvements. Process justice further concerns how different stakeholders participate in and influence TOD planning. Yip et al. [28] emphasized participation and inclusion, information sharing, power distribution, decision-making, and goal alignment among stakeholders as important dimensions of process justice in TOD. These perspectives are relevant to station-area regeneration because physical barriers, discontinuous pedestrian routes, and uneven access to commercial and public-space opportunities may have different implications for different groups. However, because the present study does not include sociodemographic distribution or institutional-process data, justice is treated as an interpretive planning consideration rather than as a directly measured outcome.

# **3 Data and Methods**

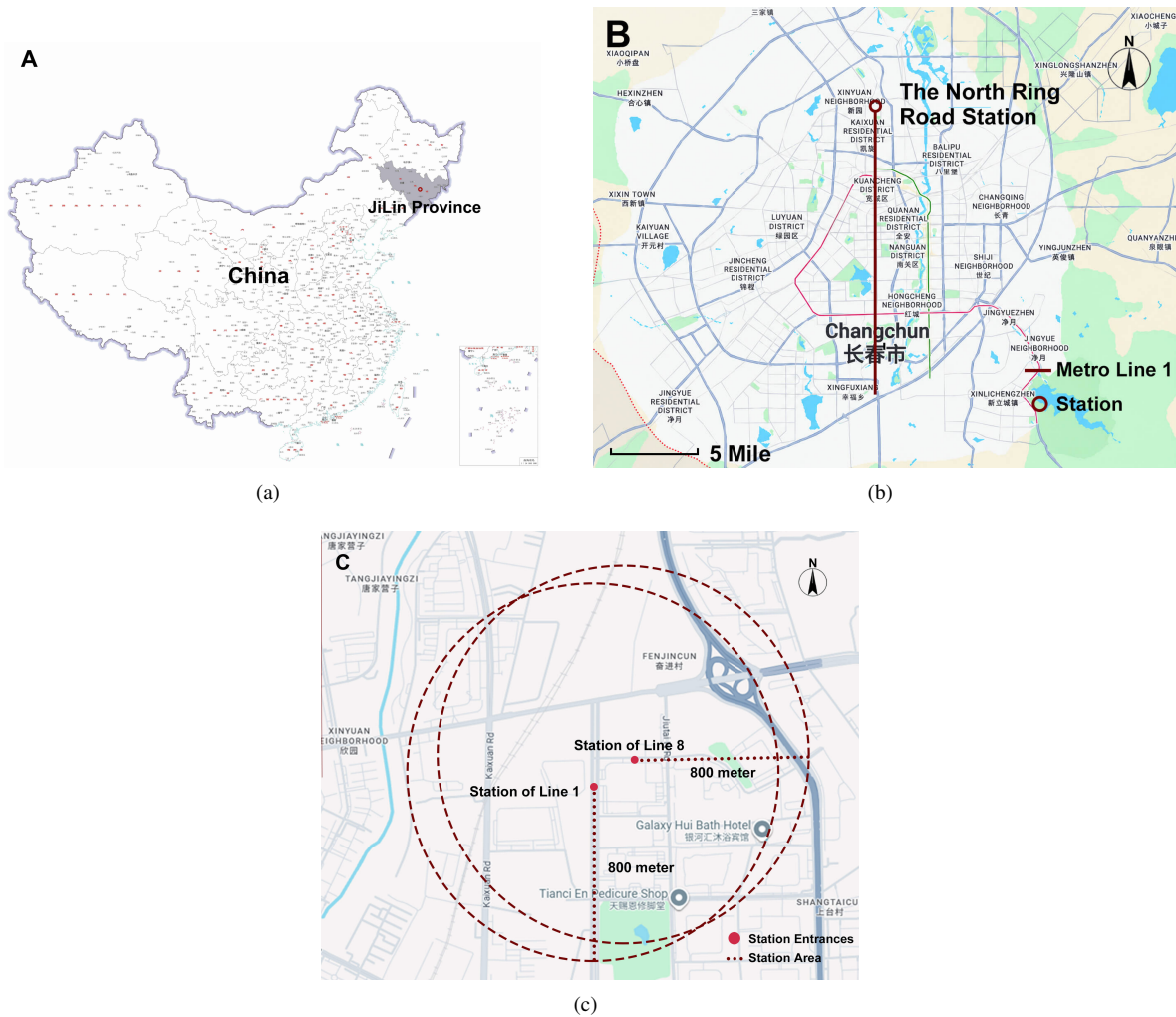
## **3.1 Study Area**

Changchun is the capital of Jilin Province and an important rail-transit city in Northeast China, as illustrated in Figure 1a. Its urban expansion has been closely linked to the growth of the rail-transit network, as discussed by Ma et al. [29]. By 2026, Changchun had 151.4 km of operating rail-transit lines. Five additional rail-transit projects were under construction, with a total construction length of 82.8 km. According to the Changchun Municipal People's Government [30], the southern extensions of Lines 1 and 3, together with the first phases of Lines 5 and 7, were

expected to open by the end of 2026, while Line 9 Phase I remained under construction. After these openings, the operating length of Changchun’s rail-transit network was expected to exceed 200 km.

As shown in Figure 1b, North Ring Road Station was selected because it is an interchange between Changchun Rail Transit Lines 1 and 8 and represents a relatively established station area with TOD-oriented and metro-related commercial development in Changchun. Its development is characterized by the coexistence of strong transit accessibility and major commercial destinations with the asynchronous development of transit infrastructure, commercial space, and the surrounding built environment. This has resulted in uneven commercial concentration, pedestrian connectivity, and observed activity within the station area. These characteristics make North Ring Road Station an informative case for examining fine-scale imbalances in TOD development. Although the station is not intended to be statistically representative of the Changchun rail-transit network, the analytical framework may provide a useful reference for examining comparable interchange-type station areas.

Following Ganning and Miller [31] and established TOD theory and station-area planning practice, an approximately 800 m area is commonly used as a conventional planning extent around a transit station, as shown in Figure 1c. This scale encompasses not only the immediate station core but also surrounding commercial, residential, and public-service areas that are functionally related to TOD. Accordingly, an 800 m TOD analytical boundary around the operational metro entrances was adopted to capture the internal differentiation of commercial spatial vitality and TOD-related built-environment conditions. The 800 m TOD analytical boundary is therefore used as a conventional planning extent for this study.



**Figure 1.** Location of the North Ring Road Station area and the 800 m transit-oriented development (TOD) analytical boundary

Source: (a) Standard map supervised by the Ministry of Natural Resources of China, Approval No. GS(2019)1651; (b–c) modified from Baidu Map.

### 3.2 Data Sources

Following Li et al. [32], the study integrated planning and land-use records, Baidu Heatmap data, commercial POIs, building footprints, road-network data, and field survey materials. All spatial datasets were projected to the World Geodetic System 1984/Universal Transverse Mercator Zone 51N (WGS 84/UTM Zone 51N) coordinate reference system and clipped to the study boundary. Commercial POIs were obtained from Amap in June 2025, extracted within the study boundary, classified according to the Amap POI Classification Standard (2023 Edition), and cleaned by removing duplicate, misclassified, and spatially inaccurate records, leaving 951 valid POIs. Amap coordinates were converted from the GCJ-02 coordinate system to WGS 84 before projection. Building data from Baidu Map and OpenStreetMap were obtained in June 2025 and checked against field observations; where measured heights were unavailable, building height was estimated from the verified number of stories using a uniform floor-to-floor height of 3 m. Road-network data from OpenStreetMap were obtained in June 2025 and checked for disconnected segments and isolated features, with broken links corrected prior to analysis, following related geospatial-data processing and quality-assessment approaches reported by Li et al. [33] and Xie et al. [34]. Field surveys conducted in June 2025 were used to verify POIs, building stories, pedestrian routes, street interfaces, and physical barriers. Table 1 summarizes the main data sources and their respective acquisition periods.

**Table 1.** Data sources used in the study

| Data                                      | Source                              | Acquisition/Observation Time                           | Main Use                                                 |
|-------------------------------------------|-------------------------------------|--------------------------------------------------------|----------------------------------------------------------|
| Planning and land-use data                | Planning documents and base maps    | Document-specific years                                | Identify functional composition and planning context     |
| Commercial points of interest (POIs) data | Amap (Gaode)                        | June 2025                                              | Analyze commercial distribution and business composition |
| Baidu Heatmap data                        | Baidu heatmap                       | 24 and 27 September 2025; 08:00–22:00 at 2-h intervals | Measure observed human activity intensity                |
| Building data                             | Baidu Map and OpenStreetMap (OSM)   | June 2025                                              | Measure development intensity and morphology             |
| Road-network data                         | OSM                                 | June 2025                                              | Derive accessibility and space-syntax indicators         |
| Field survey                              | On-site observation and photographs | June 2025                                              | Verify spatial conditions and support interpretation     |

### 3.3 Methodology

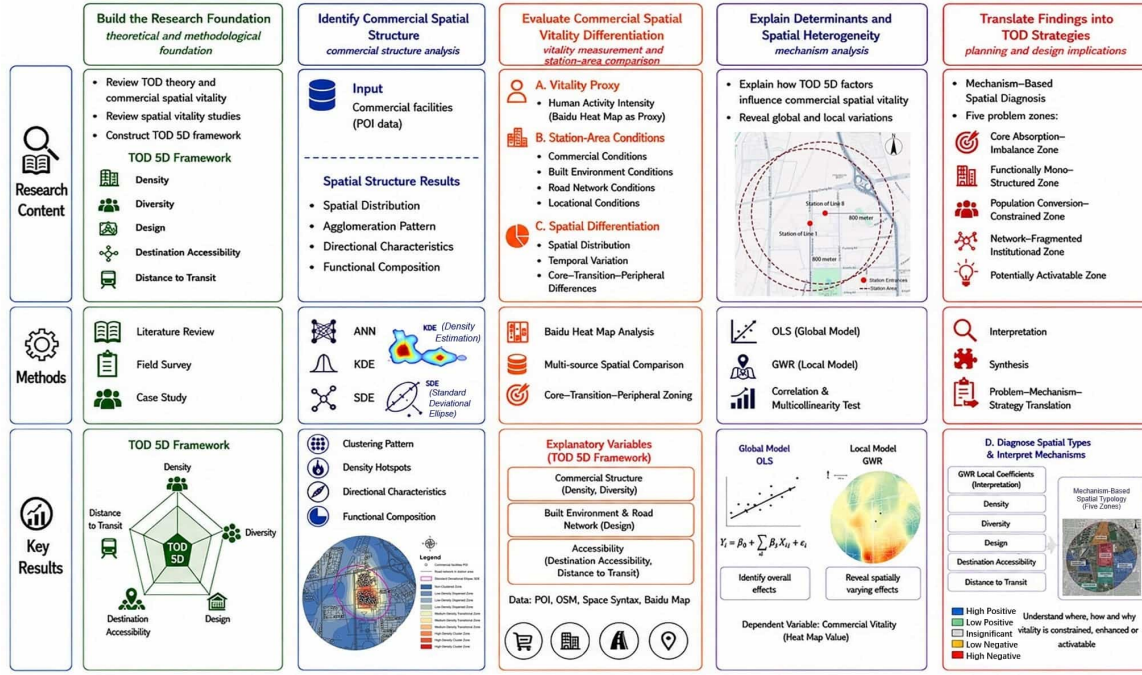
Figure 2 illustrates the analytical workflow of this study. Based on the TOD 5D framework and the spatial characteristics of the North Ring Road Station area, the processed point, polygon, line, and raster datasets were integrated into a common 30 m × 30 m analytical grid, yielding 2,704 valid grid cells.

Commercial POIs were first used to examine the spatial structure of commercial facilities. TOD-related explanatory variables were then constructed at the grid scale, while Baidu Heatmap-derived human activity intensity was used as the dependent variable, as detailed in Section 3.3.2. Spatial indicators were extracted using ArcGIS 10.8, and space-syntax indicators were calculated using DepthmapX.

OLS was used to estimate global associations between observed human activity intensity and TOD-related explanatory variables, while GWR was applied to examine spatial variation in these associations. The statistical results were subsequently interpreted together with field observations to support evidence-informed planning strategies for differentiated station-area renewal.

#### 3.3.1 Mapping the commercial structure

The first step was to characterize the spatial organization of commercial facilities within the station area. Commercial POIs were used to identify the spatial supply context against which subsequent activity patterns were interpreted. The analysis focused on three characteristics of the commercial pattern: clustering intensity, density gradient, and directional tendency. ANN was used to assess whether commercial facilities were clustered or randomly distributed. Following Lai and Wu [35], KDE was used to identify the main commercial cores and density gradients, while SDE was used to describe the main direction and spatial spread of commercial POIs. These results provided the spatial basis for interpreting subsequent observed human activity patterns.



**Figure 2.** Conceptual framework and analytical workflow for analyzing transit-oriented development (TOD)-related spatial associations in the North Ring Road Station area

### 3.3.2 Measurement of human activity intensity

Following Gao et al. [24] and Bao et al. [36], Baidu Heatmap data were used to measure observed human activity intensity at a fine spatial scale. Hereafter, this Baidu Heatmap-derived measure is referred to as “observed activity intensity”. Data were collected on 24 September 2025 (weekday) and 27 September 2025 (weekend) at two-hour intervals from 08:00 to 22:00, yielding eight observations per day and 16 observations in total. Weather conditions on both observation days were mild, with no precipitation, strong winds, or other extreme weather events likely to substantially disrupt routine outdoor activity. The two observation dates were selected to provide a weekday–weekend contrast rather than to represent seasonal or long-term activity conditions.

To ensure comparability, all heatmap images were captured using the same map extent, zoom level (18), orientation, display window, and level of detail. The images were cropped to an identical frame, georeferenced using identifiable spatial features such as road intersections, building corners, and metro entrances, and aligned to a common raster extent, cell size, and raster origin. Following Zhang [37], as shown in Figure 3, blue and light-blue areas were coded as 0 as an operational no-signal category rather than as verified observations of zero human presence. Cyan, green, yellow, orange, and red were assigned progressively higher reference activity values using the same reclassification rule for all 16 observations. The resulting seven-class maps were used only for cartographic interpretation, whereas the mean grid-level activity index derived from the 16 observations was retained as the dependent variable in the OLS and GWR models.

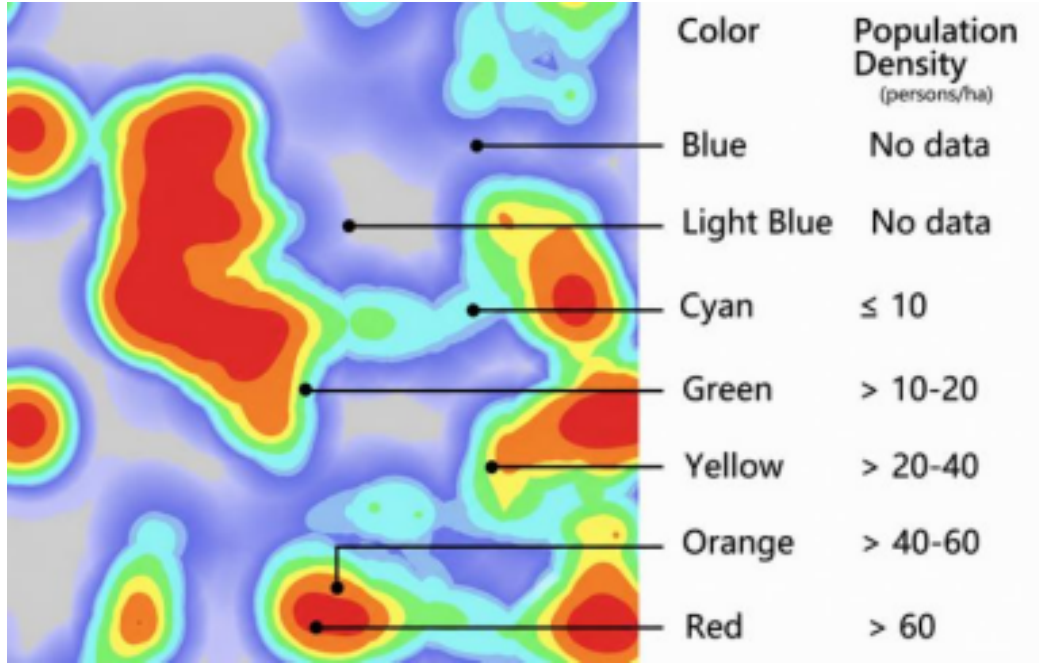
Because the public Baidu Heatmap does not provide verified individual counts or disclose its complete data-generation process, the derived values are interpreted as relative human presence and spatial concentration rather than as actual population counts, pedestrian volumes, commercial transactions, or consumer expenditure. In addition, because independent pedestrian counts or transaction data were unavailable, the results are interpreted as short-term observed weekday–weekend activity patterns rather than as indicators of persistent or long-term commercial performance.

### 3.3.3 Ordinary least squares and geographically weighted regression analysis of spatial associations

OLS and GWR were used to examine global and spatially varying associations between observed human activity intensity and TOD-related explanatory variables. OLS was first applied to estimate global associations [8]:

$$Y_i = \beta_0 + \sum_{k=1}^n \beta_k \chi_{ik} + \varepsilon_i \quad (1)$$

where,  $Y_i$  denotes the mean Baidu Heatmap-derived human activity intensity of spatial unit  $i$ ,  $\chi_{ik}$  denotes explanatory variable  $k$ , and  $\varepsilon_i$  denotes the error term. Because these associations may vary across space, GWR was used to



**Figure 3.** Reference classification of Baidu Heatmap colors for relative human activity intensity

Source: Authors' adaptation based on Zhang [37].

examine local spatial heterogeneity, following Fotheringham [12] and Fotheringham et al. [38]:

$$Y_i = \beta_0(u_i, v_i) + \sum_{k=1}^n \beta_k(u_i, v_i) \chi_{ik} + \varepsilon_i \quad (2)$$

where,  $(u_i, v_i)$  denotes the spatial coordinates of spatial unit  $i$ ;  $\beta_0(u_i, v_i)$  is the location-specific intercept;  $\beta_k(u_i, v_i)$  represents the local regression coefficient for explanatory variable  $k$  at location  $i$ ;  $\chi_{ik}$  is the value of explanatory variable  $k$  in unit  $i$ ; and  $\varepsilon_i$  is the random error term. Unlike OLS, which estimates global average associations, GWR allows regression coefficients to vary across space and therefore identifies local variation in the associations between TOD-related factors and observed activity intensity.

Before OLS and GWR estimation, all explanatory variables were transformed to the 0–1 range using Min-Max normalization, while the dependent variable retained its processed activity-intensity scale. Raster-based indicators were summarized at the grid-cell level, while vector-based indicators were assigned to the corresponding 30 m × 30 m grids through spatial extraction and spatial joins. The GWR model was estimated using a fixed bandwidth and a Gaussian kernel, with the optimal bandwidth selected by minimizing the corrected Akaike Information Criterion (AICc). The OLS adjusted coefficient of determination (adjusted  $R^2$ ) was calculated using the conventional degrees-of-freedom adjustment, while AICc was calculated from the corresponding Akaike Information Criterion (AIC) using the small-sample correction, with GWR model complexity represented by the effective number of parameters (ENP). Model validation included residual spatial-autocorrelation diagnostics, assessment of effective model complexity, and local multicollinearity diagnostics. The statistical significance of local GWR coefficients was evaluated using the Benjamini-Hochberg false discovery rate (FDR) correction at the 0.05 level. Local multicollinearity was assessed using local condition numbers and local variance inflation factors (VIFs), with condition numbers above 30 indicating potential local collinearity. Given the elevated local collinearity between distance to the nearest commercial complex (DistB) and distance to the nearest operational metro entrance (DistMetro), sensitivity analyses were additionally conducted by re-estimating the GWR model after excluding each distance variable in turn while retaining the same spatial specification.

To further assess whether the improvement in GWR performance was attributable to greater model flexibility, repeated five-fold spatial block cross-validation (hereafter Spatial-CV) was conducted. The study area was divided into 200 m × 200 m spatial blocks, with entire blocks assigned to validation folds to reduce spatial leakage. The procedure was repeated ten times. Out-of-sample predictive performance was evaluated using root mean square error (RMSE), mean absolute error (MAE), and  $R^2$ . Additional sensitivity tests were conducted using block sizes of 150, 250, and 300 m.

### 3.3.4 Transit-oriented development spatial diagnosis based on empirical evidence

Spatial diagnosis was developed from the empirical findings. This analysis integrated commercial structure, observed activity intensity, local GWR coefficients, and field observations. By jointly considering these dimensions, several diagnostic zones with different activity characteristics, development constraints, and renewal needs were delineated with reference to Vale et al. [39]. The resulting diagnosis provided the basis for formulating evidence-informed planning responses adapted to different station-area conditions, consistent with Xiao et al. [10].

## 4 Commercial Structure of the Station Area

This section examines the commercial structure of the North Ring Road Station area as the spatial context for interpreting subsequent observed activity patterns and TOD-related spatial associations.

### 4.1 Functional Structure and Major Commercial Anchors

The North Ring Road Station area has a mixed land-use structure, dominated by residential (26.20%), commercial (17.82%), educational (10.17%), and administrative (6.50%) functions. Its commercial system is organized around two major anchors (Figure 4a): a consumption-oriented southern core centered on the Sasseur Outlets and Dictator Park complex (Figures 4b, 4c), and a northern wholesale-logistics core anchored by the Hefa Building Materials Market (Figure 4d). This dual-anchor structure provides the spatial context for interpreting subsequent spatial variation in observed activity intensity, consistent with Wang et al. [26].

### 4.2 Spatial Clustering and Concentric Distribution of Commercial Facilities

The analysis identified 951 commercial POIs within the 800 m TOD analytical boundary. Following the ANN approach of Clark and Evans [11], the analysis confirmed that commercial POIs are not randomly distributed but exhibit a clear clustering tendency within the study area (Figure 5). The KDE result further identified the main areas of commercial concentration (Figure 6). Commercial density is highest around the station core and the southern commercial cluster. From these areas, commercial facilities extend along major roads, especially North Ring Road. Some small clusters also appear in peripheral residential areas. These clusters are mainly related to daily services rather than the outward spread of the main commercial core. This pattern shows that the station area has both centralized commercial cores and localized service nodes, consistent with Tu et al. [23] and Gao et al. [24].

The SDE in Figure 6 shows a clear directional pattern. Commercial facilities are arranged along the corridor between the northern and southern clusters, rather than spreading evenly around the metro station. Following Yuill [40], this result suggests that the commercial structure is corridor-oriented. The concentric-layer results in Figure 7 show a clear distance-decay pattern. The 0–300 m core contains 612 POIs, accounting for 64.35% of the total, with a density of 1,538/km<sup>2</sup>. The density falls to 167/km<sup>2</sup> in the 300–500 m zone and then rises slightly to 180/km<sup>2</sup> in the 500–800 m zone. This slight rebound does not indicate the outward expansion of the main commercial core. It is more closely related to localized community services in peripheral residential areas. Therefore, the station area shows a concentrated core structure, while the outer areas are mainly characterized by neighborhood-serving commercial uses.

### 4.3 Business Composition and Diversity of Commercial Facilities

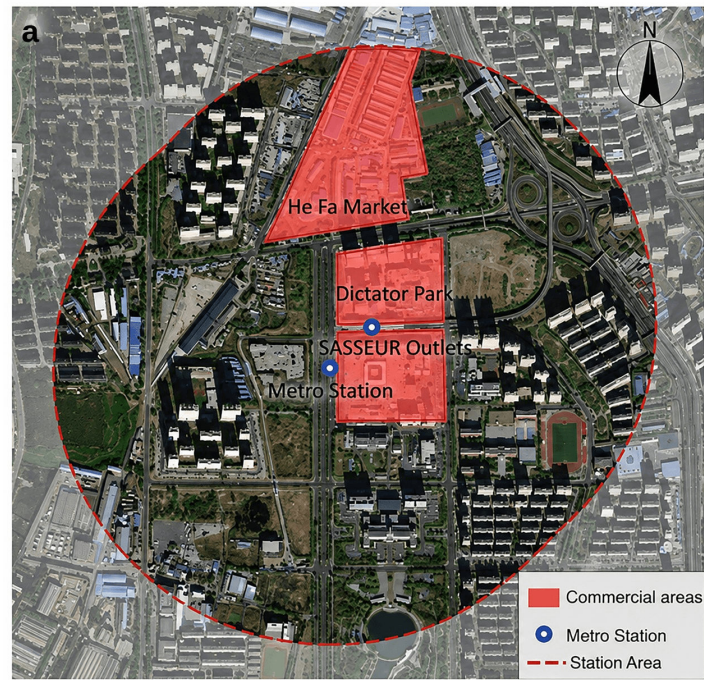
As shown in Table 2, retail (44.06%) and food-and-beverage (25.13%) account for the largest shares of the station-area commercial system. Retail clusters are concentrated in the station core and southern center; food-and-beverage establishments follow major corridors; and daily services and medical care are more dispersed in peripheral areas, consistent with Liu et al. [41]. Commercial diversity is concentrated in the southern core and along main thoroughfares, whereas specialized sites, such as the wholesale market, exhibit high commercial concentration but relatively low diversity, consistent with He and Zhang [42] and Zhuo et al. [43].

## 5 Observed Activity Intensity and Explanatory Variables

Following Gao et al. [24], this section uses Baidu Heatmap-derived human activity intensity as a spatial proxy for the observable activity dimension of commercial spatial vitality in the North Ring Road Station area and treats commercial structure, built-environment conditions, and locational accessibility as explanatory factors in the subsequent OLS and GWR analyses.

### 5.1 Spatial Pattern of Observed Activity Intensity

Observed activity intensity was derived from the processed Baidu Heatmap data at the grid-cell level, following the method described in Section 3.3.2. Here, the focus shifts from data processing to its spatial pattern. Observed activity is primarily concentrated in the southern retail core area and near the metro station (Figures 8 and 9). This area has high observed activity intensity and a high concentration of commercial functions. Activity intensity extends



(a)



(b)

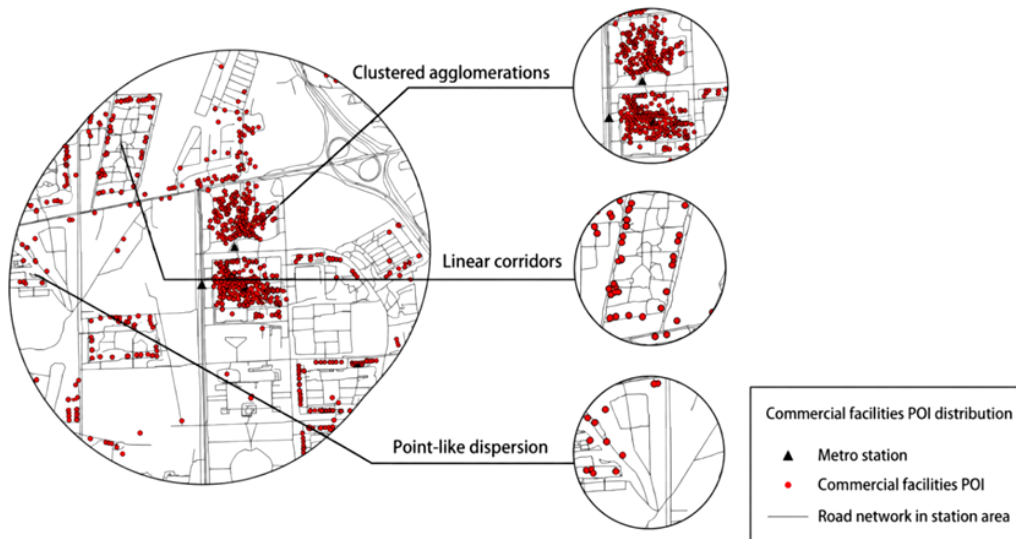


(c)

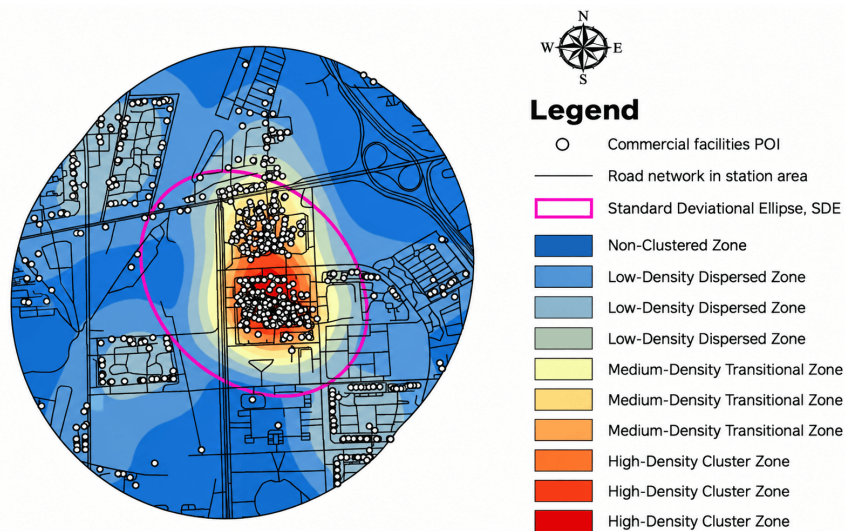


(d)

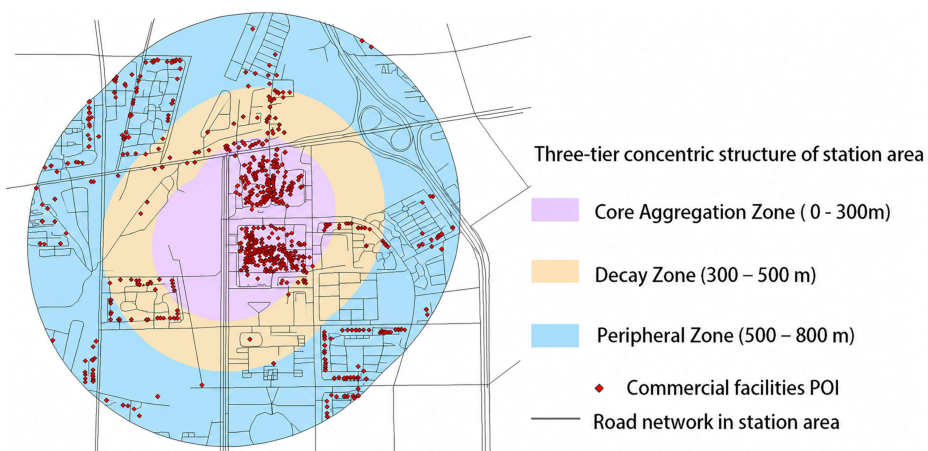
**Figure 4.** Functional structure and major commercial anchors of the station area: (a) land-use structure of the station area; (b) Sasseur Outlets Commercial Complex; (c) Dictator Park Commercial Complex; (d) Hefa Building Materials Market



**Figure 5.** Distribution of commercial points of interest (POIs) across different spatial layers



**Figure 6.** The standard deviational ellipse (SDE) and kernel density estimation (KDE) of commercial points of interest (POIs)



**Figure 7.** Distribution of commercial points of interest (POIs) across different concentric zones

**Table 2.** Classification and proportional distribution of commercial points of interest (POIs) within the station area

| Category                    | Number of Facilities (POIs) | Proportion |
|-----------------------------|-----------------------------|------------|
| Retail commerce             | 419                         | 44.06%     |
| Food & beverage             | 239                         | 25.13%     |
| Daily services              | 133                         | 13.99%     |
| Sports & recreation         | 50                          | 5.26%      |
| Science/culture & education | 49                          | 5.15%      |
| Medical services            | 37                          | 3.89%      |
| Accommodation               | 24                          | 2.52%      |
| Total                       | 951                         | 100%       |

from here along the North Ring Road and Jiutai North Road. The overall pattern is corridor-like rather than evenly distributed outward from the station center, consistent with Qi et al. [44].

Several moderately active areas emerge in the surrounding residential areas. These active areas coincide mainly with community-oriented commercial and daily-service facilities and do not represent outward expansion from the main commercial core area. Observed activity intensity in the North Ring Road Station area exhibits characteristics of core concentration, corridor extension, and limited outward rebound.

## 5.2 Indicator Selection Based on the Transit-Oriented Development 5D Framework

Drawing on Yu et al. [9] and Porta et al. [45], the indicator system was constructed according to the TOD 5D framework (Table 3), covering Density, Diversity, Design, Destination Accessibility, and Distance to Transit. Density was operationalized through commercial facility concentration and building development intensity because these indicators characterize the concentration of commercial activity opportunities and the horizontal and vertical intensity of the built environment. Diversity was represented by the Shannon Diversity Index, which captures the richness and balance of commercial functions. Design was represented by road-network density and space-syntax indicators because these variables characterize street configuration, local accessibility, connectivity, topological depth, and potential through-movement. Destination Accessibility was represented by geometric proximity to major commercial-complex entrances, which serve as important non-transit activity anchors, whereas Distance to Transit was represented by DistMetro, defined as the Euclidean distance to the nearest operational metro entrance. Thus, the two distance variables capture proximity to different types of activity anchors, while road-network and space-syntax variables separately represent the configurational properties of the street network. Field-observed interface openness, pedestrian permeability, barriers, and frontage conditions were used only for contextual interpretation and spatial diagnosis and were not entered as quantitative predictors in the OLS or GWR models.

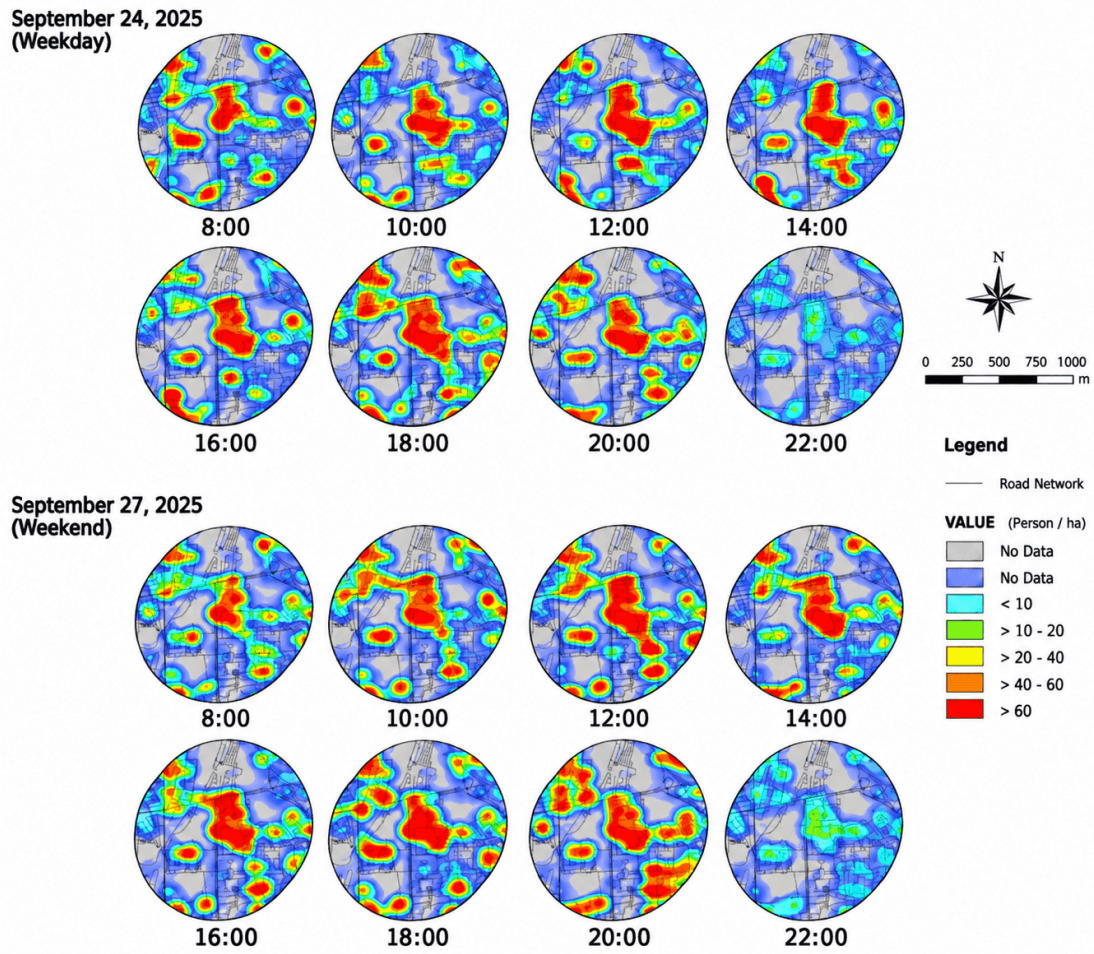
## 6 Spatial Associations and Local Heterogeneity

Before estimating the models, the explanatory variables were tested for multicollinearity, and spatial dependence in observed activity intensity was examined. The VIF values ranged from 1.34 to 6.78, with a mean of 2.95, indicating no severe multicollinearity. Following Moran [46], Global Moran's I was significant at the 0.01 level, indicating positive spatial autocorrelation in observed activity intensity. Areas with similar activity-intensity levels tended to form spatial clusters. This pattern indicates that spatial dependence should be considered in subsequent model assessment.

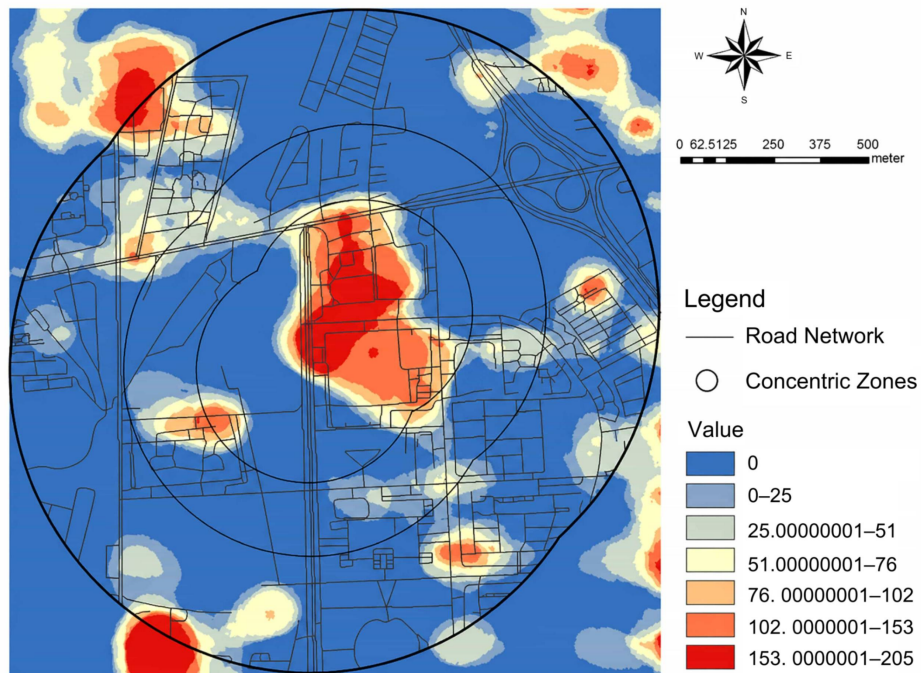
OLS was first used as a global benchmark to identify average associations between observed activity intensity and TOD-related explanatory variables. Moran's I was then applied to the OLS residuals to test whether spatially structured errors remained. The residual diagnostic results were used to assess the adequacy of the global model and to determine the need for further spatially explicit analysis. GWR was subsequently applied to estimate spatially varying coefficients and examine local heterogeneity in built-environment and locational associations.

### 6.1 Global Associations Identified by Ordinary Least Squares

The OLS model provides the global baseline for interpreting observed activity intensity (Figure 10). Commercial Facility Kernel Density shows the strongest positive association ( $\beta = 178.98, p < 0.001$ ), and Shannon Diversity of Commercial Facilities is also significantly positive ( $\beta = 59.88, p < 0.001$ ), indicating positive associations with commercial agglomeration and functional diversity, consistent with Duan and Chen [47]. Morphological, network, and locational factors show mixed associations. Building density is negatively associated with observed activity intensity ( $\beta = -20.62, p < 0.001$ ). Among the road-network variables, Road Network Kernel Density and Road Choice are positively related to observed activity intensity, whereas Local Integration (HH.R3) and Road



**Figure 8.** Baidu heatmap patterns of relative human activity intensity in the station area



**Figure 9.** Aggregated spatial pattern of observed human activity intensity in the station area

**Table 3.** Operationalization of observed human activity intensity and transit-oriented development (TOD) 5D indicators

| Category                            | Dimension                          | Indicator                                            | TOD 5D Dimension          | Calculation Method                                                                | Description                                                    |
|-------------------------------------|------------------------------------|------------------------------------------------------|---------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------|
| Activity intensity                  | Population activity representation | Observed activity representation                     | –                         | $H_i = \frac{1}{n} \sum_{t=1}^n H_{it}$                                           | Mean heatmap-derived activity intensity across 16 observations |
| Commercial facility characteristics | Commercial facility condition      | Commercial facility kernel density                   | Density                   | $D(\chi) = \frac{1}{nh} \sum_{i=1}^N K\left(\frac{d(\chi, \chi_i)}{h}\right)$     | Local concentration of commercial points of interest (POIs)    |
|                                     |                                    | Shannon diversity of commercial facilities           | Diversity                 | $H = - \sum_{i=1}^n p_i \ln(p_i)$                                                 | Commercial functional diversity                                |
|                                     |                                    | Building density                                     | Density                   | $BD_i = \frac{B_i^{area}}{A_i}$                                                   | Building footprint ratio within each grid                      |
|                                     |                                    | Building height                                      | Density                   | –                                                                                 | Mean building height estimated using 3 m per story             |
| Built environment characteristics   | Spatial morphology                 | Road network kernel density                          | Design                    | $D(\chi) = \frac{1}{nh} \sum_{i=1}^n L_i K\left(\frac{d(\chi, \chi_i)}{h}\right)$ | Local road-network concentration                               |
|                                     |                                    | Road depth                                           | Design                    | $MD_i = \frac{\sum_{j=1}^k d_{ij}}{k-1}$                                          | Topological depth; lower values indicate higher accessibility  |
|                                     |                                    | Hillier-Hanson local integration at radius 3 (HH_R3) | Design                    | $R_{n,i} = \frac{2(MD_i-1)}{k-2}$                                                 | Local topological integration within a three-step radius       |
|                                     |                                    | Road connectivity                                    | Design                    | $Connectivity_i = \sum_{j=1}^k \delta_{ij}$                                       | Number of directly connected segments                          |
|                                     |                                    | Road choice                                          | Design                    | $Choice(i) = \sum_{s \neq t \neq j} \frac{\sigma_{st}(i)}{\sigma_{st}}$           | Through-movement potential                                     |
|                                     |                                    | Road betweenness                                     | Design                    | $BC(i) = \sum_{s \neq t \neq i} \frac{\sigma_{st}(i)}{\sigma_{st}}$               | Intermediary role in origin-destination paths                  |
| –                                   | Location conditions                | Distance to the nearest commercial complex           | Destination accessibility | $D_{ij} = \sqrt{(\chi_i - \chi_j)^2 + (y_i - y_j)^2}$                             | Distance to the nearest major commercial anchor                |
|                                     |                                    | Distance to the nearest operational metro entrance   | Distance to transit       |                                                                                   | Euclidean distance to the nearest operational metro entrance   |

Note: An en dash indicates “not applicable” or “no data”. Field-observed conditions were used for contextual interpretation and spatial diagnosis but were not included as quantitative predictors in the ordinary least squares (OLS) or geographically weighted regression (GWR) models.

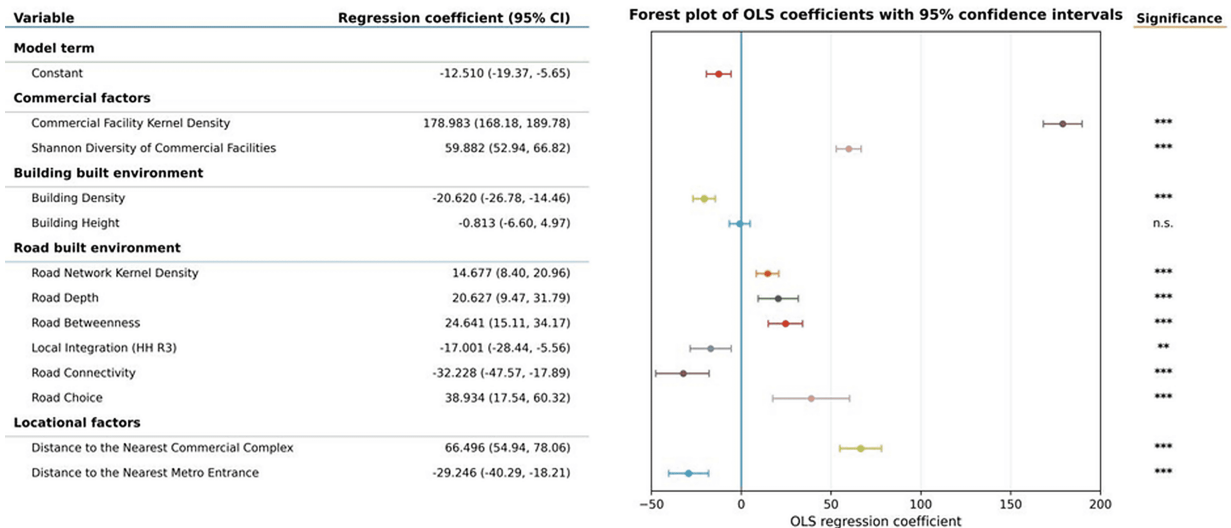
Connectivity show negative associations. These contrasting coefficients indicate that different dimensions of street-network configuration are associated with observed activity intensity in different ways, consistent with Sevtsuk [48]. Distance from the nearest operational metro entrance shows a significant negative association with observed activity intensity ( $\beta = -29.25$ ,  $p < 0.001$ ). Areas farther from the station generally exhibit lower activity levels. The distance coefficient to the nearest commercial complex is positive. This suggests that observed activity is not confined to locations closest to major commercial complexes. The OLS residuals showed significant positive spatial autocorrelation (Moran’s I = 0.917,  $Z = 66.962$ ,  $p < 0.001$ ), indicating that substantial spatial dependence remained after global model estimation and that the global model did not fully capture the spatial structure of observed activity, thereby motivating further examination of spatially varying associations.

## 6.2 Spatial Heterogeneity Identified by Geographically Weighted Regression

As shown in Table 4,  $R^2$  increased from 0.595 to 0.884, while AICc, AIC, Bayesian information criterion (BIC), mean squared error (MSE), RMSE, and MAE all decreased, indicating a substantial improvement in in-sample model fit and supporting further examination of spatially varying associations.

Residual spatial autocorrelation remained significant in both models, although Moran’s I decreased from 0.917 for OLS to 0.797 for GWR (both  $p < 0.001$ ). This indicates that GWR captured part of the spatial structure remaining in the OLS residuals but did not eliminate spatial dependence. The AICc-selected fixed Gaussian bandwidth was 146.7 m, with an ENP of 146.114. Accordingly, the GWR results indicate spatially varying associations while some spatial structure remains unexplained. The model is therefore used to diagnose local variation rather than to represent the complete underlying spatial process.

Local multicollinearity was then examined before interpreting the individual local coefficients. Local condition



**Figure 10.** Coefficient plot of ordinary least squares (OLS) regression results for observed human activity intensity

numbers ranged from 35.76 to 3343.04 (median = 270.23), with all grid cells exceeding the conventional threshold of 30. Local VIFs were particularly high for DistB and DistMetro, with median values of 52.46 and 53.32, respectively. Sensitivity analyses excluding either distance variable substantially reduced local collinearity while leaving the principal associations of commercial concentration, functional diversity, and road-network conditions broadly stable. The median local condition number decreased to 62.62 without DistB and 56.45 without DistMetro, while the median VIF of the remaining distance variable fell below 4. DistB and DistMetro were therefore major contributors to local collinearity, although some collinearity remained after either variable was excluded. Their individual local coefficients are consequently interpreted cautiously as related locational-accessibility gradients, because the high local condition numbers indicate that some local coefficient estimates may be unstable, particularly those of DistB and DistMetro. The coefficient maps of DistB and DistMetro are therefore treated as exploratory and are not used independently to support specific planning recommendations.

To determine whether the apparent improvement of GWR was confined to in-sample fit or reflected greater model flexibility, repeated spatial block cross-validation provided additional out-of-sample evidence for GWR relative to OLS. Across ten repeated five-fold validations, the mean RMSE decreased from 30.529 to 27.045 and the mean MAE from 21.857 to 18.232, while mean out-of-sample  $R^2$  increased from 0.518 to 0.621. Similar performance advantages were observed with spatial block sizes ranging from 150 to 300 m, indicating that the improved predictive performance of GWR was not confined to in-sample goodness-of-fit or solely attributable to greater model flexibility. Nevertheless, although GWR improved model fit and spatial cross-validation performance, significant residual spatial autocorrelation remained, indicating that it did not fully account for the spatial structure of observed activity intensity. The local coefficients are therefore interpreted as diagnostic spatial associations rather than as a complete representation of the underlying spatial process or as causal relationships. Table 5 summarizes the distributions of the estimated local coefficients and illustrates the extent of spatial variation across the study area.

The statistical significance of local GWR coefficients was further evaluated using the FDR correction. Table 6 reports the proportions of significant positive, significant negative, and non-significant coefficients.

The FDR-adjusted results show that Commercial Facility Kernel Density and Shannon Diversity have the largest proportions of significant positive local associations, whereas DistMetro has a comparatively large proportion of significant negative associations. The coexistence of positive, negative, and non-significant local coefficients further supports spatial variation in TOD-related associations with observed activity intensity. Given the elevated local collinearity of DistB and DistMetro, their individual coefficients are interpreted cautiously as exploratory locational-accessibility gradients.

The spatial distributions of these coefficients (Figure 11), interpreted together with the FDR-adjusted significance results, further reveal distinct local association patterns. Commercial Facility Kernel Density shows the strongest positive association at the interface between the southern retail cluster and residential areas but weakens in the northern wholesale cluster, where specialization and inward-facing layouts coincide with lower observed activity intensity. Shannon Diversity is positively associated with activity in mixed-use transitional corridors but remains weak in isolated enclaves. Road Network Kernel Density is positively associated with activity in residential belts but negatively associated along traffic-dominated segments with physical severance, while Road Choice shows stronger positive associations where through-movement coincides with permeable street edges, consistent with Istrate [49].

**Table 4.** Comparison of model performance between OLS and GWR

| Metric                 | OLS                | GWR                |
|------------------------|--------------------|--------------------|
| Adjusted $R^2$         | 0.593              | –                  |
| AIC                    | 25716.37           | 22610.91           |
| AICc                   | 25716.51           | 22627.96           |
| BIC                    | 25793.10           | 23479.25           |
| MSE                    | 782.90             | 224.83             |
| RMSE                   | 27.98              | 14.99              |
| MAE                    | 19.92              | 10.10              |
| Selected bandwidth (m) | –                  | 146.7              |
| ENP                    | –                  | 146.114            |
| Residual Moran's I     | 0.917              | 0.797              |
| Residual $p$ -value    | <0.001             | <0.001             |
| Spatial-CV RMSE        | 30.529 $\pm$ 0.318 | 27.045 $\pm$ 1.279 |
| Spatial-CV MAE         | 21.857 $\pm$ 0.277 | 18.232 $\pm$ 0.697 |
| Spatial-CV $R^2$       | 0.518 $\pm$ 0.010  | 0.621 $\pm$ 0.036  |

Note: OLS—ordinary least squares; GWR—geographically weighted regression;  $R^2$ —coefficient of determination; AIC—Akaike information criterion; AICc—corrected Akaike information criterion; BIC—Bayesian information criterion; MSE—mean squared error; RMSE—root mean square error; MAE—mean absolute error; ENP—effective number of parameters; Spatial-CV—spatial cross-validation. An en dash indicates “not applicable” or “no data”. Spatial-CV values are reported as means  $\pm$  standard deviations from ten repeated five-fold spatial block cross-validations using 200 m  $\times$  200 m spatial blocks. Adjusted  $R^2$  is reported for OLS only. For GWR, model complexity is represented by ENP and AICc because the effective degrees of freedom depend on the selected bandwidth.

The association with DistMetro is generally negative, although positive associations occur in some residential clusters, while Building Height shows positive associations mainly in eastern transition zones. Overall, TOD-related variables exhibit spatially varying associations across local socio-spatial contexts, consistent with Liu [50].

Taken together, these diagnostics support the use of GWR as a diagnostic tool for describing spatially varying associations in the study area. However, the remaining residual spatial autocorrelation and local multicollinearity indicate that the model should not be interpreted as a complete or causal representation of the underlying spatial process.

## 7 Discussion

This section discusses the principal findings of the study by integrating the observed commercial structure and activity patterns with the global and spatially varying associations identified by the OLS and GWR analyses. It further considers justice and stakeholder implications and translates the empirical diagnosis into evidence-informed TOD-based planning responses.

### 7.1 Interpretation of the Main Findings

The findings reveal a spatially uneven relationship among commercial structure, observed activity intensity, and TOD-related spatial conditions within the North Ring Road Station area. The commercial structure exhibits a dual-core and corridor-oriented pattern, characterized by a consumption-oriented southern core and a specialized northern wholesale-logistics core. Observed activity intensity broadly corresponds to this spatial structure, with the strongest activity concentrated around the southern retail core and metro station, corridor-like extension along major roads, and localized moderate activity in surrounding residential areas. These patterns indicate that commercial concentration and observed activity are spatially related but do not diffuse uniformly throughout the station area.

Overall, observed activity intensity in the station area is associated with multiple TOD-related spatial conditions, with transit proximity representing only one component, consistent with Xiao et al. [10]. Globally, Commercial Facility Kernel Density and Shannon Diversity of Commercial Facilities are positively associated with observed activity intensity, indicating the relevance of commercial supply and functional mix. Morphological, network, and locational variables show more spatially differentiated associations.

The contrasting associations among Road Network Kernel Density, Road Choice, and Local Integration (HH\_R3) indicate that different dimensions of street-network configuration are associated with observed activity intensity in

**Table 5.** GWR estimation results for observed human activity intensity in the North Ring Road Station area

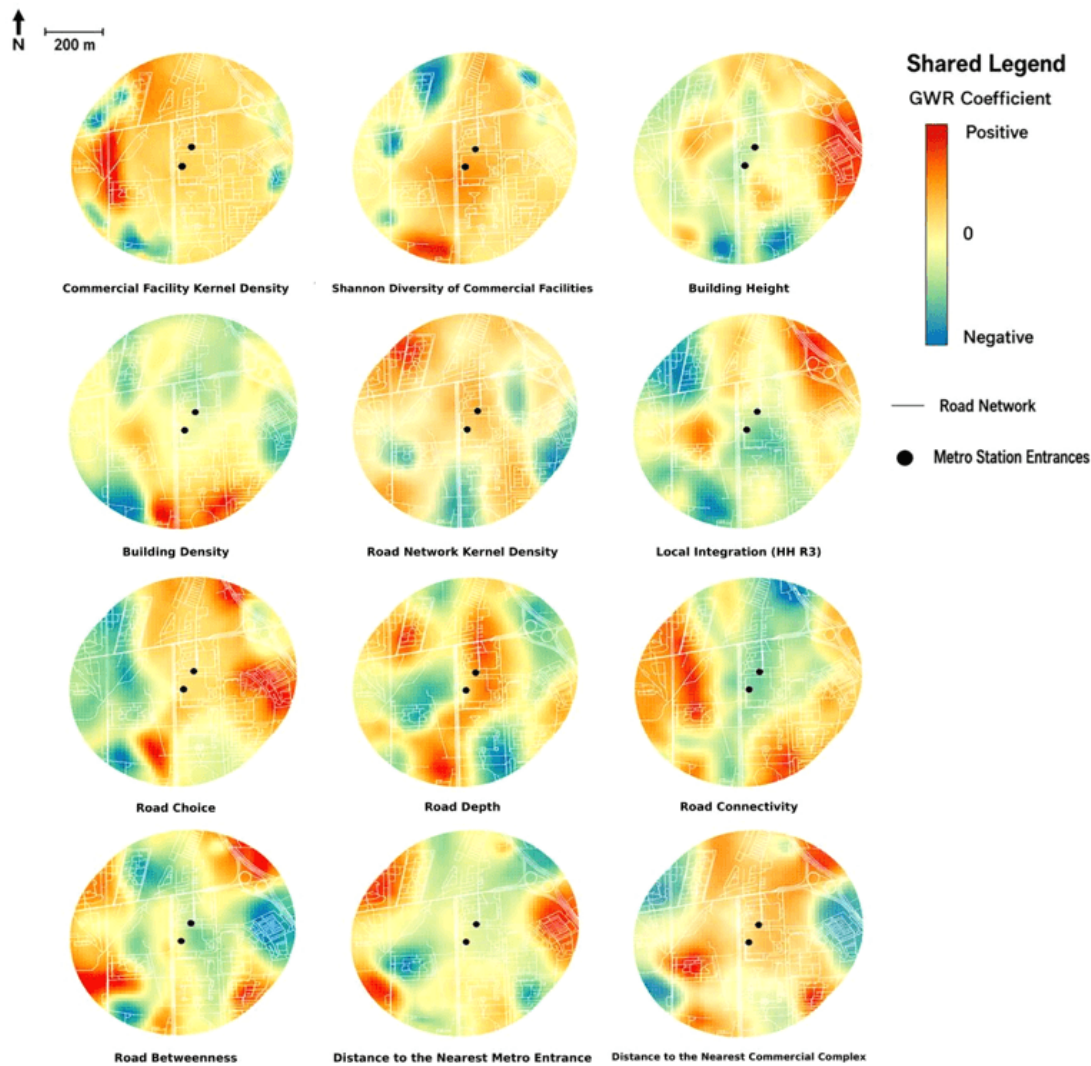
| Variable                                           | Variable Code | Min       | Q1       | Median  | Q3      | Max     | Mean    | Std. Dev. |
|----------------------------------------------------|---------------|-----------|----------|---------|---------|---------|---------|-----------|
| Constant                                           | Constant      | −167.042  | −42.542  | −12.968 | 16.383  | 132.446 | −13.217 | 51.727    |
| Commercial facility kernel density                 | B_kernel      | −480.036  | 41.469   | 147.547 | 215.519 | 854.461 | 146.965 | 193.246   |
| Shannon diversity of commercial facilities         | SHDI          | −46.244   | 10.630   | 32.322  | 67.423  | 224.311 | 41.485  | 43.448    |
| Building density                                   | BuDens        | −157.001  | −35.135  | −11.323 | 14.853  | 262.106 | −0.769  | 58.185    |
| Building height                                    | BuHeight      | −233.029  | −28.224  | −4.081  | 21.417  | 126.392 | −3.805  | 48.674    |
| Road network kernel density                        | Roaddens      | −74.639   | −1.982   | 19.010  | 48.924  | 207.449 | 24.367  | 42.883    |
| Road depth                                         | Depth         | −157.367  | −36.337  | −2.214  | 35.951  | 147.593 | −1.261  | 53.712    |
| Local integration (HH.R3)                          | HH.R3         | −168.931  | −42.999  | −8.669  | 11.363  | 128.856 | −16.219 | 49.465    |
| Road connectivity                                  | R3            | −188.648  | −37.235  | 6.994   | 56.795  | 143.933 | 7.189   | 62.263    |
| Road choice                                        | Choice        | −773.624  | −100.966 | 29.996  | 68.322  | 250.014 | −23.912 | 149.146   |
| Road betweenness                                   | Betweenness   | −121.715  | −10.276  | 17.525  | 48.144  | 117.848 | 16.900  | 42.587    |
| Distance to the nearest commercial complex         | DistB         | −1065.084 | −94.082  | 111.084 | 247.830 | 812.408 | 27.817  | 349.739   |
| Distance to the nearest operational metro entrance | DistMetro     | −822.461  | −158.785 | −66.416 | 102.457 | 997.093 | 3.523   | 308.167   |

Note: GWR—geographically weighted regression; HH.R3—Hillier–Hanson local integration at radius 3; SHDI—Shannon diversity index; Q1—first quartile; Q3—third quartile; Std. Dev.—standard deviation.

**Table 6.** FDR-adjusted significance distribution of local GWR coefficients

| Variable    | Significant Positive (%) | Significant Negative (%) | Non-significant (%) |
|-------------|--------------------------|--------------------------|---------------------|
| B_kemel     | 69.9                     | 6.5                      | 23.6                |
| SHDI        | 65.7                     | 5.3                      | 29.1                |
| BuDens      | 23.8                     | 41.6                     | 34.7                |
| BuHeight    | 29.9                     | 37.1                     | 33.0                |
| Roaddens    | 53.3                     | 16.9                     | 29.8                |
| Depth       | 28.3                     | 28.9                     | 42.8                |
| HH.R3       | 12.8                     | 32.8                     | 54.4                |
| R3          | 28.8                     | 26.5                     | 44.6                |
| Choice      | 43.4                     | 27.2                     | 29.4                |
| Betweenness | 43.0                     | 14.9                     | 42.1                |
| DistB       | 56.9                     | 25.5                     | 17.6                |
| DistMetro   | 28.5                     | 51.8                     | 19.7                |

Note: FDR—false discovery rate; GWR—geographically weighted regression; B\_kernel—commercial facility kernel density; SHDI—Shannon diversity index; BuDens—building density; BuHeight—building height; Roaddens—road network kernel density; HH.R3—Hillier–Hanson local integration at radius 3; R3—road connectivity; DistB—distance to the nearest commercial complex; DistMetro—distance to the nearest operational metro entrance.



**Figure 11.** Spatial distribution of GWR local coefficients for multidimensional explanatory variables

Note: GWR—geographically weighted regression; HH\_R3—Hillier–Hanson local integration at radius 3.

different ways, consistent with Qi et al. [44]. These findings should be interpreted as spatial associations rather than causal effects. The GWR results further show that the same TOD-related factor may have positive associations in one subarea and weaker or negative associations in another, consistent with previous studies [50–52]. In the Chinese urban regeneration context, the Changchun case shows that high transit accessibility can coexist with uneven street-level activity. Field observations further suggest that the spatial relationship between transit accessibility and street-level activity may be conditioned by interface permeability, functional mix, and pedestrian-scale continuity. These findings provide a useful reference for comparable transit-led station areas where rail investment has preceded fine-grained station-area regeneration.

Although GWR showed improved in-sample and out-of-sample performance relative to OLS, the remaining residual spatial autocorrelation and local multicollinearity indicate that the local coefficients should continue to be interpreted cautiously as diagnostic associations rather than causal effects.

## 7.2 Justice and Stakeholder Implications

These findings also have implications for spatial justice in metro-oriented regeneration. The observed coexistence of short Euclidean distances to metro entrances with wide arterials, gated boundaries, discontinuous pedestrian routes, and uneven access to commercial and public-space opportunities suggests that nominal transit proximity does not necessarily imply equally convenient access across all users or subareas. This interpretation is consistent with Tzanni et al. [27], who highlighted the social dimensions of spatial justice in public-transport use. However, because the present study does not include sociodemographic characteristics, individual travel behavior, or distributional outcomes, it cannot determine whether particular social groups experience disproportionate benefits or burdens.

Spatial justice is therefore used here as a planning lens for interpreting uneven accessibility conditions rather than as an empirically measured outcome.

Process justice is likewise relevant to translating the study findings into regeneration action. As Yip et al. [28] emphasize, participation and inclusion, information sharing, power distribution, decision-making, and goal alignment shape the fairness of TOD processes. Accordingly, implementation of pedestrian, interface, and land-use interventions should involve municipal and district authorities, metro operators, commercial property owners, small businesses, community organizations, residents, pedestrians, and transit users, particularly where priorities and trade-offs differ across diagnostic zones. The stakeholder implications below therefore concern not only who may benefit from the findings, but also who should be involved in identifying priorities and shaping locally appropriate interventions.

The findings also have practical implications for stakeholders involved in station-area planning, transport operation, commercial development, and everyday use. Different stakeholders may use the spatial diagnosis and proposed planning responses according to their respective responsibilities and interests. Table 7 summarizes the main stakeholder groups and the potential applications and benefits of the study findings.

Building on the preceding discussion and stakeholder implications, the OLS and GWR associations, observed activity patterns, commercial spatial structure, and field observations are translated into evidence-informed TOD-based planning responses for enhancing commercial spatial vitality. The combined evidence indicates that the spatial continuity of observed activity is associated with the coordination of transit accessibility, commercial functions, pedestrian connectivity, and street-interface conditions.

### **7.3 Evidence-Informed Diagnosis of Barriers to Activity Continuity**

The GWR results indicate substantial spatial variation in the associations between TOD-related variables and observed activity intensity. Combined with field observations, three spatial conditions are consistently associated with discontinuities between transit accessibility, commercial spaces, and observed street-level activity. First, field-observed physical severance near wide arterials and overpasses coincides with activity discontinuities despite short Euclidean distances to the station, resulting in a “visually proximate but behaviorally distant” condition (Figures 12a, 12b), consistent with Ewing and Dumbaugh [53]. Second, the northern wholesale cluster combines high Commercial Facility Kernel Density with comparatively weaker positive local associations and field-observed inward-facing or inactive interfaces (Figure 12c). This mismatch suggests that commercial concentration alone does not correspond to equally strong street-level activity. Third, field-observed gated walls and construction barriers coincide with breaks in pedestrian routes and weaker network continuity, limiting connections between commercial interfaces and surrounding neighborhoods (Figure 12d), consistent with Páez et al. [54]. Together, these observations indicate that spatial continuity, interface openness, pedestrian connectivity, and transit–commercial connections are important contextual conditions associated with variations in observed activity intensity.

### **7.4 Evidence-Informed Spatial Typology for Commercial Vitality Enhancement**

The GWR results were interpreted together with observed activity intensity, commercial density and diversity, TOD 5D indicators, land-use characteristics, and field-observed spatial conditions. Based on this integrated evidence, five diagnostic zones were delineated to represent dominant local constraints and planning conditions (Figure 13 and Table 8). These zones are evidence-informed planning categories rather than outputs of formal statistical clustering. Their boundaries are therefore interpretive boundaries for planning diagnosis rather than precise statistical classifications.

The five diagnostic zones (Figure 13) show that similar levels of transit proximity do not necessarily correspond to similar activity patterns. In the station core and around major commercial destinations, strong activity and commercial concentration are not always accompanied by continuous connections to surrounding streets. In other areas, major roads, gated boundaries, and enclosed interfaces coincide with weaker pedestrian connections, while peripheral and transitional areas may contain local demand and spatial resources but lack sufficient functional diversity, destination accessibility, or pedestrian support. These differences provide the spatial basis for the differentiated planning responses discussed below.

### **7.5 Transit-Oriented Development-Based Strategies for Commercial Spatial Vitality Enhancement**

The proposed strategies are planning hypotheses informed by statistically observed associations and field-observed spatial constraints. They should not be interpreted as demonstrated intervention effects because the proposed spatial and design measures were not directly evaluated through experimental, longitudinal, or before–after analysis. Their role is therefore to translate the identified evidence–diagnosis relationships into context-specific directions for station-area renewal.

In the Core Absorption-Imbalance Zone, high activity and commercial concentration coexist with enclosed frontages, large building blocks, and road barriers that weaken connections with surrounding public spaces. Street-

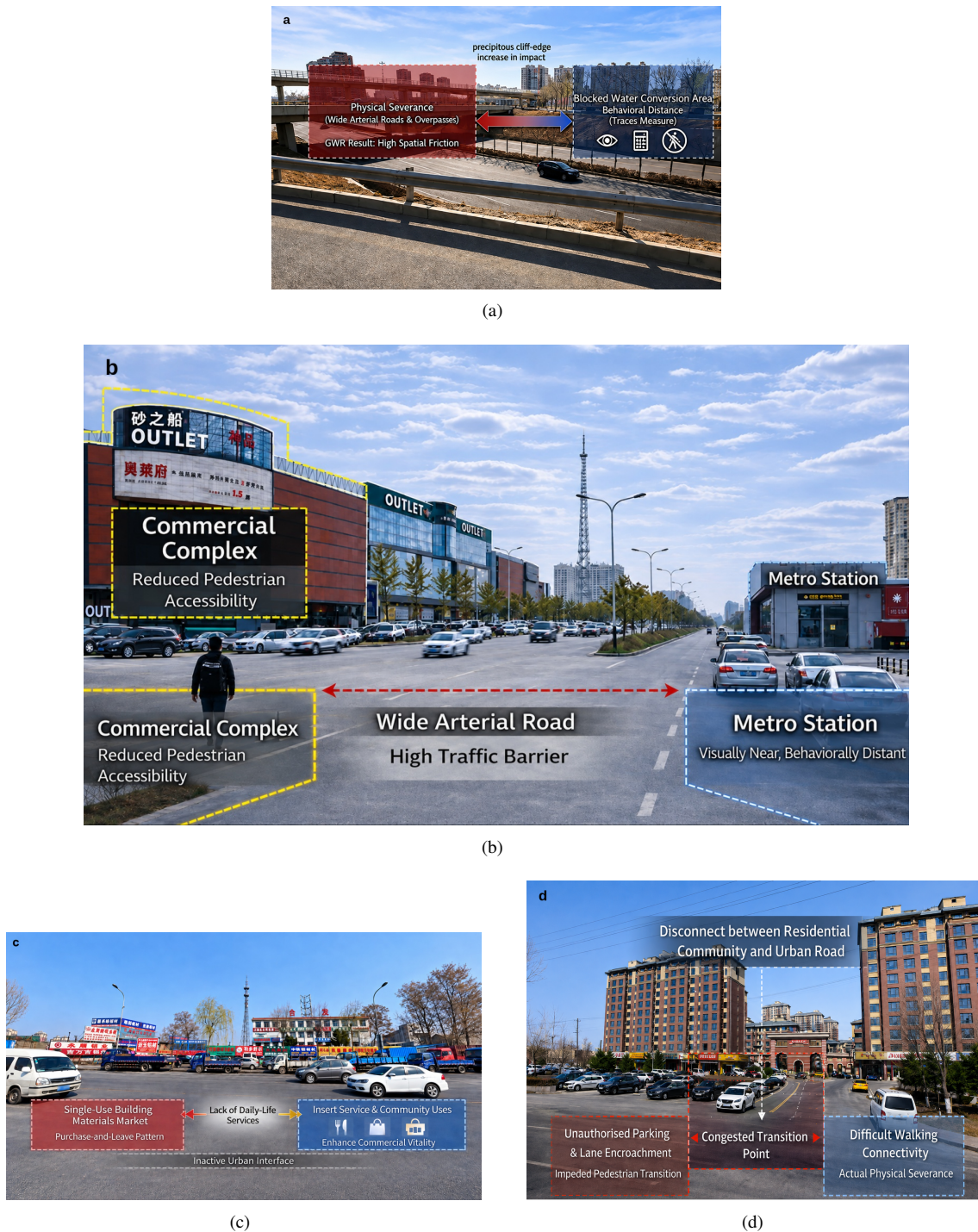
**Table 7.** Stakeholders and potential applications of the study findings

| Stakeholder                                              | Potential Applications and Benefits                                                                                                                                             |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Municipal planning and urban-regeneration authorities    | Identify priority regeneration areas and match interventions to locally identified spatial constraints                                                                          |
| Metro and transport authorities                          | Improve metro-entrance accessibility, pedestrian transfer routes, and first/last-mile connections                                                                               |
| Commercial property owners and market operators          | Improve frontage permeability, functional diversity, and connections between major commercial anchors and surrounding streets                                                   |
| Small businesses and local retailers                     | May benefit from improved pedestrian continuity and stronger activity spillover from major commercial destinations                                                              |
| Residents, pedestrians, and transit users                | May benefit from safer crossings, shorter walking routes, and improved access to transit, commercial services, and public spaces                                                |
| District and community-level authorities                 | Support incremental and locally responsive improvements in residential and transitional areas                                                                                   |
| Community organizations and neighborhood representatives | Contribute local knowledge on pedestrian barriers and everyday access needs, participate in prioritizing interventions, and support more inclusive station-area decision-making |

**Table 8.** Evidence–diagnosis–planning response matrix for commercial spatial vitality enhancement

| Diagnostic Zone                   | Key Evidence                                                                                  | Diagnosis                                                                     | Planning Response                                                            |
|-----------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Core absorption imbalance         | High activity and commercial concentration; inward-facing interfaces and road barriers        | Strong core concentration but weak street continuity                          | Improve pedestrian links, frontage permeability, and public-space interfaces |
| Functionally monostructured       | High commercial concentration; low diversity; specialized and inactive interfaces             | Concentration without corresponding street-level activity                     | Add complementary services, leisure uses, and active frontages               |
| Population-conversion constrained | Spatially varying design/distance-to-transit associations; wide roads and gated edges         | Weak connection between transit accessibility and surrounding street activity | Improve crossings, pedestrian links, entrances, and boundary permeability    |
| Destination-diffusion constrained | Heterogeneous locational accessibility associations; discontinuous links around major anchors | Uneven spatial diffusion from major commercial destinations                   | Strengthen direct walking routes, secondary nodes, and commercial corridors  |
| Potentially activatable           | Moderate activity; mixed design/diversity associations; underused spaces                      | Spatial resources exist but functional and pedestrian support remain weak     | Introduce small activity nodes, flexible uses, and public-space improvements |

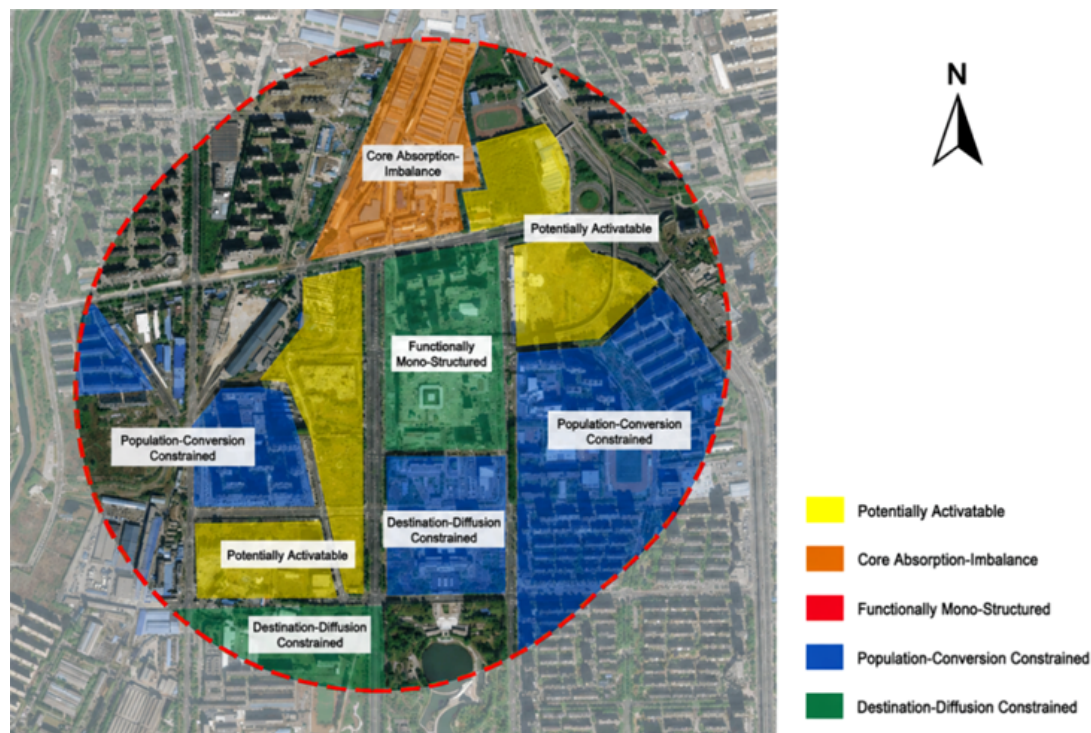
Note: The diagnostic-zone boundaries are interpretive planning boundaries derived from the combined evidence above rather than precise statistical classifications.



**Figure 12.** Typical field-observed physical barriers associated with activity discontinuity: (a) physical severance by overpasses; (b) behavioral barriers caused by wide arterial roads; (c) inactive interfaces at single-use markets; (d) topological disconnection at residential-road transitions

facing entrances, permeable ground-floor interfaces, safer crossings, and clearer pedestrian connections between metro entrances and nearby commercial spaces are therefore proposed as context-specific planning responses, informed by Hu et al. [55] and Nikolopoulou and Steemers [56]. Similar measures are relevant to the Destination-Diffusion-Constrained Zone, where direct walking routes, secondary commercial or service nodes, and first/last-mile connections could strengthen links between major destinations, residential areas, and intermediate streets. Recent spatial planning research has also demonstrated the value of integrating accessibility, traffic demand, and proximity to commercial and transport nodes in location-specific transport infrastructure planning [57].

In the Functionally Mono-Structured Zone, particularly the northern wholesale area, high commercial density



**Figure 13.** Evidence-informed spatial typology of five diagnostic zones in the North Ring Road Station area

is accompanied by limited functional diversity and purpose-specific visits. Rather than increasing development intensity, renewal should introduce complementary daily services, convenience retail, food and beverage facilities, and community-oriented functions that connect the market with surrounding neighborhoods. Evening and weekend uses may further extend activity beyond the operating periods of specialized commerce, consistent with Carmona [58].

In the Population-Conversion-Constrained Zone, a short Euclidean distance to the station does not necessarily correspond to convenient pedestrian access. Wide arterials, gated boundaries, construction barriers, and incomplete pedestrian links interrupt connections among residential areas, metro entrances, and street-level shops. Improvements should therefore focus on safer crossings, shorter detours, feasible boundary openings, clearer pedestrian connections to metro entrances, and small public spaces at key junctions, consistent with Southworth [59]. These measures also support transport integration by strengthening pedestrian transfer and first/last-mile connections, although actual passenger-flow conversion was not measured.

The Potentially Activatable Zones require more gradual intervention. Temporary commercial uses, small community facilities, improved walking connections, and modest public-space upgrades could support the gradual activation of underused or transitional spaces while responding to nearby residential demand.

## 7.6 Spatial Heterogeneity and the Transit-Oriented Development 5D Framework

The GWR results provide a spatially differentiated interpretation of the TOD 5D framework. Within the station area, density, diversity, design, destination accessibility, and distance to transit show varying associations with observed activity intensity. Design and Distance to Transit are particularly relevant where pedestrian barriers and fragmented routes constrain effective station access, while diversity is positively associated with activity in many mixed-use areas, consistent with Xiao et al. [10]. These findings indicate that high density or transit accessibility alone does not necessarily correspond to strong street-level activity where spatial permeability and interface openness are limited. TOD-based renewal should therefore coordinate metro-entrance access, pedestrian transfer routes, first/last-mile connections, and adjacent commercial and land-use functions according to local spatial conditions; however, actual passenger-flow conversion was not directly measured in this study. Accordingly, the study evaluates the spatial conditions that may support transport–place integration rather than directly estimating metro passenger-flow conversion.

## 8 Conclusions

This study examined commercial spatial vitality in the North Ring Road Station area from a TOD perspective, using Baidu Heatmap-derived human activity intensity as a spatial proxy for its observable activity dimension. Commercial concentration and functional diversity are positively associated with observed activity intensity,

while built form, road-network structure, destination accessibility, and distance to transit show spatially varying associations. The dual-core and corridor-oriented pattern further indicates uneven activity distribution, highlighting the need to coordinate metro access, pedestrian connections, commercial interfaces, and surrounding land uses.

At the micro scale, the TOD 5D framework demonstrates diagnostic value for identifying local spatial heterogeneity. TOD-related conditions should therefore be interpreted together with pedestrian, functional, and interface characteristics. Accordingly, measures such as interface activation, functional integration, pedestrian connectivity, and localized activity nodes are proposed as evidence-informed planning responses based on statistical associations and field observations.

Although the empirical findings are derived from a single interchange station in Changchun, the proposed analytical framework may be applied to other station areas as a benchmarking approach rather than as a fixed set of planning prescriptions. Comparative application may be guided by several questions: Is observed activity concentrated around the station or spatially continuous along surrounding pedestrian routes? Do commercial concentration and functional diversity correspond to stronger street-level activity? Are metro entrances and major destinations connected by direct and permeable pedestrian routes? Do major commercial anchors generate activity spillover into surrounding streets? Which TOD 5D dimensions exhibit the strongest local spatial variation? These questions can support comparison across different station types, land-use contexts, governance systems, and urban environments. Accordingly, the benchmarking approach may support comparative application in metro station areas worldwide, provided that differences in local planning institutions, governance arrangements, mobility behavior, development intensity, and data availability are explicitly considered. The broader applicability of this study therefore lies primarily in its fine-scale diagnosis-to-strategy framework rather than in the direct transfer of specific coefficients or interventions.

Several limitations should be noted. Baidu Heatmap data reflect relative human presence and short-term activity patterns rather than commercial transactions, consumer behavior, or actual metro passenger flows. The 800 m TOD analytical boundary represents a conventional planning extent and does not fully capture variations in effective pedestrian accessibility associated with physical barriers, while the single-station design limits direct generalization. Nevertheless, the analytical framework integrating fine-scale activity measurement, TOD 5D analysis, spatial diagnosis, and the translation of findings into planning responses may be transferable to comparable station areas. Spatial block cross-validation provided additional out-of-sample support for GWR relative to OLS, although significant residual spatial dependence remained, indicating that GWR did not fully account for the spatial structure of observed activity intensity. The local coefficients should therefore be interpreted as diagnostic spatial associations rather than as a complete spatial process or causal relationships, particularly for the two distance-related indicators affected by local collinearity. Future research should compare multiple station types and incorporate longer-term activity data and network-based accessibility measures.

### **Author Contributions**

Conceptualization, D.W.Z. and A.N.R.; methodology, D.W.Z. and A.N.R.; software, D.W.Z.; formal analysis, D.W.Z.; investigation and data curation, D.W.Z.; visualization, D.W.Z.; writing—original draft preparation, D.W.Z.; writing—review and editing, D.W.Z. and A.N.R.; supervision, A.N.R.; project administration, D.W.Z. In addition, D.W.Z. conducted the empirical data processing and spatial analyses, while A.N.R. provided methodological guidance and supervisory review. All authors have read and agreed to the published version of the manuscript.

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### **Data Availability**

The data supporting the findings of this study are available from the corresponding author upon reasonable request. Publicly accessible road-network data were obtained from OpenStreetMap. Other datasets, including commercial POIs data, Baidu heatmap data, Baidu Map building data, planning and land-use records, and field survey materials, are subject to the access conditions and terms of use of the respective data providers and are therefore not publicly deposited in a repository.

### **Conflicts of Interest**

The authors declare no conflicts of interest.

## Declaration on the Use of Generative AI and AI-assisted Technologies

During the preparation of this manuscript, the authors used generative AI and AI-assisted technologies for language polishing, grammar checking, and formatting improvement. The authors reviewed and edited all AI-assisted outputs and take full responsibility for the content of the manuscript. No generative AI or AI-assisted technologies were used to fabricate data, results, references, or to replace the authors' intellectual contributions.

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